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Digital Logic Design Project

3-Mode Digital Timer

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1 Introduction

This project involves the design and implementation of a multifunctional Digital Timer capable of operating in three distinct modes. The system is built using discrete logic components from the 74xx TTL family, demonstrating core digital electronics concepts such as counting, decoding, comparison, and data multiplexing.

2 Hardware Components

- **74192:** Synchronous Up/Down BCD Counter.
- **7447:** BCD to 7-Segment Decoder/Driver.
- **7485:** 4-bit Magnitude Comparator.
- **74157:** Quad 2-to-1 Data Selector/Multiplexer.
- **7473:** Dual J-K Flip-Flop with Clear.
- **7-Segment Display:** Common Anode.

3 System Operation Modes

3.1 Normal Timer Mode

In this mode, the **74192** counters receive a clock signal (e.g., 1Hz) to count upwards or downwards. The BCD output is sent directly to the **7447** decoders to be displayed on the 7-segment units.

3.2 Alarm Set Mode

The user can preset a specific time value. The **74157** multiplexer switches the inputs of the comparator to the manual switches. The **7485** comparator continuously checks the counter's value against the preset value. When they match, a signal is triggered.

3.3 Second Adjustment Mode

A specialized mode that allows the user to increment or decrement the "Seconds" counter independently.

4 Logical Design Implementation

4.1 Counting and Cascading

The **74192** counters are cascaded to handle two digits. The Terminal Count output of the first stage triggers the next stage.

4.2 Comparison and Feedback

The **7473** Flip-Flop stores the alarm state. When the **7485** detects equality, it activates the alarm until reset.

5 Conclusion

The 3-mode Digital Timer successfully demonstrates the practical application of combinational and sequential logic.

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